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PULLING JIG, OPTICAL CABLE PULLING METHOD, AND OPTICAL CABLE PULLING SYSTEM

Abstract

A pulling jig for pulling a plurality of optical cables while maintaining a state in which the optical cables are arranged in parallel, the pulling jig includes a base portion having a width equal to or more than a width in the state in which the plurality of optical cables are arranged in parallel, a pulling portion for enabling the base portion to be connected to a pulling device, and a plurality of connecting portions provided at the base portion and arranged in a width direction along the width. The connecting portions are connectable to the optical cables in at least one-to-one correspondence.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a pulling jig, an optical cable pulling method, and an optical cable pulling system. This application claims priority based on Japanese Patent Application No. 2024-032397 filed on Mar. 4, 2024, and the entire contents of the Japanese patent application are incorporated herein by reference.

BACKGROUND

[0002] Japanese Unexamined Patent Application Publication No. 2023-083096 discloses that a server rack group including a plurality of server racks and a distributing frame are connected by an optical cable.

SUMMARY

[0003] A pulling jig according to an aspect of the present disclosure is a pulling jig for pulling a plurality of optical cables while maintaining a state in which the optical cables are arranged in parallel. The pulling jig includes a base portion having a width equal to or more than a width in the state in which the plurality of optical cables are arranged in parallel, a pulling portion for enabling the base portion to be connected to a pulling device, and a plurality of connecting portions provided at the base portion and arranged in a width direction along the width. The connecting portions are connectable to the optical cables in at least one-to-one correspondence.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. **1** illustrates an optical communication system according to the present embodiment.

[0005] FIG. **2** is a schematic diagram of an optical branch cable.

[0006] FIG. **3** is a cross-sectional view of a main cable.

[0007] FIG. **4** is a partial enlarged view of a tape.

[0008] FIG. **5** illustrates a branch cord bag.

[0009] FIG. **6** is a cross-sectional view of the VI-VI cross section in FIG. **2**.

[0010] FIG. **7** is a perspective view of a bending restriction member.

[0011] FIG. **8** illustrates a bending restriction member and a branch cord portion.

[0012] FIG. **9** illustrates an installation of an optical branch cable using a pulling jig.

[0013] FIG. **10** is a plan view of a pulling jig.

[0014] FIG. **11** is a cross-sectional view of the XI-XI plane in FIG. **10**.

[0015] FIG. **12** illustrates a fixed position of a connecting portion.

[0016] FIG. **13** is a diagram of an optical branch cable as viewed of a direction intersecting a width direction.

[0017] FIG. **14** illustrates an optical branch cable aligned by a pulling jig.

[0018] FIG. **15** is a perspective view of an optical branch cable set according to a modification.

[0019] FIG. **16** illustrates an optical branch cable set and a branch cord bag according to the modification.

[0020] FIG. **17** illustrates an arrangement example of a branch cord portion of each optical branch cable.

DETAILED DESCRIPTION

[0021] Depending on the number of server racks arranged in the server rack group, a plurality of optical cables may be installed in parallel so as to extend in the same direction. The work of installing many optical cables has been complicated.

[0022] An object of the present disclosure is to improve the installation workability when installing a plurality of optical cables.

Description of Embodiments of Present Disclosure

[0023] First, embodiments of the present disclosure will be listed and described.

[0024] (1) A pulling jig according to an aspect of the present disclosure is a pulling jig for pulling a plurality of optical cables while maintaining a state in which the optical cables are arranged in parallel. The pulling jig includes a base portion having a width equal to or more than a width in the state in which the plurality of optical cables are arranged in parallel, a pulling portion for enabling the base portion to be connected to a pulling device, and a plurality of connecting portions provided at the base portion and arranged in a width direction along the width. The connecting portions are connectable to the optical cables in at least one-to-one correspondence.

[0025] The pulling jig includes a plurality of connecting portions arranged along the width direction, and each connecting portions is configured to be connectable to one optical cable, allowing the plurality of optical cables to be pulled with each one connected to the connecting portions. This enables the plurality of optical cables to be pulled at once with as equal force as possible, the installation workability of the optical cables are improved.

[0026] (2) In the pulling jig according to (1), each of the optical cables may include a main cable including a plurality of optical fiber core wires, and a branch cable portion in which at least one of the optical fiber core wires is led out from the main cable. The connecting portions may be fixable at any one of at least two different positions in a direction intersecting the width direction.

[0027] Depending on the optical cable, the branching portion where the main cable and the branch cable portion branch off may be formed at a position shifted from the end portion of the optical cable.

[0028] According to the pulling jig, the connecting portion is fixable in one of two different positions in a direction intersecting the width direction, even when the position of the branching portion is shifted in the longitudinal direction depending on the optical cable, the positions of the branching portions in the longitudinal direction are easily aligned to some extent by the displacements of the connecting portions.

[0029] (3) In the pulling jig according to (2), a guide portion extending in the direction intersecting the width direction may be formed at the base portion. Each of the connecting portions may be displaceable along the guide portion.

[0030] According to the pulling jig, since the connecting portion is configured to be displaced along the guide portion, a mechanism capable of changing the position of the connecting portion in the direction intersecting the width direction can be realized by a simple configuration.

[0031] (4) In the pulling jig according to (3), the guide portion may be formed as a guide slit. Each of the connecting portions may include a positioning portion and an engaging tool. The positioning portion may include a shaft portion being able to be inserted through the guide slit, an external thread being formed on at least a portion of an outer surface, and an engaging portion disposed below the guide slit, the engaging portion being not able to be inserted through the guide slit. An internal thread being able to be fastened to the shaft portion may be formed in the engaging tool. A position of the connecting portion may be fixed with respect to the base portion in the direction intersecting the width direction by the engaging tool being fastened to the shaft portion.

[0032] According to the pulling jig, the engaging tool is fastened to the shaft portion, and thus it is possible to realize a mechanism for fixing the movement and the position of the connecting portion with respect to the base portion in the direction intersecting the width direction with a simple configuration.

[0033] (5) An optical cable pulling method according to an aspect of the present disclosure includes pulling a plurality of optical cables while maintaining a state in which the optical cables are arranged in parallel using the pulling jig according to any one of (1) to (4).

[0034] According to the optical cable pulling method, since the plurality of optical cables can be pulled at the same time by using the pulling jig, the installation workability of the optical cables are improved.

[0035] (6) An optical cable pulling system according to an aspect of the present disclosure includes the pulling jig according to any one of (1) to (4), and an optical cable set including a plurality of optical cables. Each of the optical cables includes a main cable including a plurality of optical fiber core wires, and a branch cable portion in which at least one of the optical fiber core wires is branched from the main cable. The plurality of optical cables are arranged in parallel in the width direction. The connecting portions are connected to the optical cables in one-to-one correspondence.

[0036] According to the optical cable pulling system, since one optical cable is connected to each of the connecting portions, it is possible to pull a plurality of optical cables simultaneously with as equal force as possible. Thus, since the plurality of optical cables can be pulled simultaneously, the installation workability of the optical cables are improved.

Details of Embodiments of Present Disclosure

[0037] Specific examples of a pulling jig, an optical cable pulling method, and an optical cable pulling system according to embodiments of the present disclosure will be described below with reference to the drawings. The present disclosure is not limited to these illustrations, but is defined by the appended claims, and is intended to include all modifications within the scope and meaning equivalent to the appended claims.

Optical Branch Cable

[0038] FIG. **1** illustrates an optical communication system **100** according to the present embodiment. As illustrated in FIG. **1**, optical communication system **100** includes a distributing frame **110** and a plurality of server racks **120**. Optical branch cables **1** used in optical communication system **100** are connected to distributing frame **110**. Server rack **120** accommodates physical servers therein. An optical fiber branched from optical branch cable **1** is connected to each server rack **120**. Optical branch cable **1** is installed on a cable laying tray T and wired to distributing frame **110** and server rack **120**.

[0039] FIG. **2** is a schematic view of optical branch cable **1**. As illustrated in FIG. **2**, optical branch cable **1** includes a main cable **10**, a branch cord portion **20**, a branch cord bag **30**, a bending restriction member **40**, an end portion bag **50**, and a connecting loop **60**.

[0040] FIG. **3** is a cross-sectional view of main cable **10**. Main cable **10** includes a plurality of optical cords **11**, a tension member **12**, and a tape **13**. Optical cord **11**, tension member **12**, and tape **13** are arranged along the longitudinal direction of main cable **10**.

[0041] Each of optical cords **11** includes at least one optical fiber core wire **11**A therein. In the present embodiment, optical cord **11** includes 24 optical fiber core wires **11**A therein. The plurality of optical cords **11** are bundled in a first bundle **14** to constitute an optical cord group **15**.

[0042] Tension member **12** is formed of, for example, fiber reinforced plastic (FRP) such as aramid FRP, glass FRP, or carbon FRP. Tension member **12** is configured to be applied with tension when optical branch cable **1** is pulled. Optical cord group **15** and tension member **12** are bundled by a second bundle **16** to constitute a core portion **17**.

[0043] Tape **13** forms the outermost layer of main cable **10**. Tape **13** is configured to maintain a shape covering core portion **17** by being wound around core portion **17** by longitudinal wrapping. Tape **13** is a sheet-like insulating resin, and may be formed of, for example, polyester.

[0044] Returning to FIG. **2**, branch cord portion **20** is a cord in which at least one optical fiber core wire **11**A is led out from optical cord **11** of main cable **10** to the outer side of tape **13** which is the outermost layer. An optical connector **21** is provided at the end of branch cord portion **20**. Branch cord portion **20** may be a portion in which optical fiber core wire **11**A in optical cord **11** is led out to the outer side of the outermost layer of main cable **10**, or may be a portion in which optical cord **11** itself is led out to the outer side of the outermost layer of main cable **10**.

[0045] FIG. **4** is a partially enlarged view of tape **13**. As illustrated in FIG. **4**, tape **13** is wound around core portion **17** by longitudinal wrapping, so that an overlapping portion is formed where tape **13** overlaps in double layers. In the overlapping portion, an end portion of the inner layer portion of tape **13** in the circumferential direction is referred to as a first end portion **13**A, and an end portion of the outer layer portion of tape **13** in the circumferential direction is referred to as a second end portion **13**B.

[0046] Tape **13** has a slit S formed at a position corresponding to the branching portion at which branch cord portion **20** is branched from main cable **10**. Slit S is formed in a portion of tape **13** in the circumferential direction so as to extend in the circumferential direction of main cable **10**.

[0047] Slit S is formed to penetrate the inner layer portion and the outer layer portion of the overlapping portion. Slit S is formed from first end portion **13**A to second end portion **13**B in the overlapping portions of the longitudinal wrapping of tape **13**. In other words, slit S is formed to include first end portion **13**A and second end portion **13**B.

[0048] Branch cord portion **20** is led out from main cable **10** through slit S. FIG. **4** illustrates branch cord portion **20** passing through slit S by an arrow. It is noted that, the dashed line indicates that branch cord portion **20** passes through the inner side of tape **13** which is the outermost layer of main cable **10**.

[0049] As illustrated in FIG. **4**, branch cord portion **20** passes through the inner side of the outermost layer formed by tape **13** before branching from main cable **10**. Branch cord portion **20** is led out from main cable **10** by passing through slit S. After being branched, branch cord portion **20** is arranged to the outer side of the outermost layer.

Branch Cord Bag

[0050] FIG. **5** illustrates branch cord bag **30**. In FIG. **5**, the outer shape line of branch cord bag **30** is shown by a dashed line. FIG. **6** is a cross-sectional view of the VI-VI cross section in FIG. **2**.

[0051] Branch cord bag **30** is configured to be capable of housing branch cord portion **20**. Branch cord portion **20** is accommodated in branch cord bag **30** when it is not necessary to connect to server rack **120** (refer to FIG. **1**) during transportation or the like. Branch cord bag **30** is configured to be capable of housing branch cord portion **20** in a state where at least a portion of branch cord portion **20** is wound in a coil shape.

[0052] Branch cord bag **30** has a flat shape, with a portion of it being formed to be openable. At least one surface of branch cord bag **30** may have a semicircular shape. In the present embodiment, the outer surface of branch cord bag **30** may have two main surfaces **31**. Main surface **31** is a surface having the largest area in branch cord bag **30**. Main surface **31** of branch cord bag **30** may have a semicircular shape so as to follow branch cord portion **20** wound in a coil shape.

[0053] As illustrated in FIG. **6**, a first hook or loop fastener portion **32***a* and a second hook or loop fastener portion **32***b* are formed on the inner surface of branch cord bag **30**. First hook or loop fastener portion **32***a* and second hook or loop fastener portion **32***b* are formed at positions of main surface **31** where the opening of branch cord bag **30** is formed so that the coupling surfaces thereof face main cable **10**. In the present embodiment, first hook or loop fastener portion **32***a* and second hook or loop fastener portion **32***b* form a hook surface of the hook and loop fastener.

[0054] A cable hook or loop fastener portion **18** is formed on the outer surface of main cable **10**. Cable hook or loop fastener portion **18** may be provided only at a position corresponding to the branching portion. In the present embodiment, cable hook or loop fastener portion **18** forms a loop surface.

[0055] First hook or loop fastener portion **32***a* and second hook or loop fastener portion **32***b* are configured to be capable of coupling to cable hook or loop fastener portion **18**. In other words, the inner surface of branch cord bag **30** and main cable **10** are coupled by first hook or loop fastener portion **32***a*, second hook or loop fastener portion **32***b*, and cable hook or loop fastener portion **18**. In the present embodiment, a third hook or loop fastener portion **32***c* forms a hook surface of the hook and loop fastener.

[0056] As illustrated in FIG. **6**, third hook or loop fastener portion **32***c* is formed on the inner surface of branch cord bag **30**. Third hook or loop fastener portion **32***c* is formed to protrude in the direction of the opening of branch cord bag **30**.

[0057] On the outer surface of main surface **31** where second hook or loop fastener portion **32***b* is provided in branch cord bag **30**, a fourth hook or loop fastener portion **32***d* is formed. In the present embodiment, fourth hook or loop fastener portion **32***d* forms a loop surface.

[0058] Third hook or loop fastener portion **32***c* is configured to be capable of coupling to fourth hook or loop fastener portion **32***d*. As illustrated in FIG. **6**, third hook or loop fastener portion **32***c* is coupled to fourth hook or loop fastener portion **32***d* by performing a coupling operation O**1** to bend the portion provided with third hook or loop fastener portion **32***c* onto fourth hook or loop fastener portion **32***d*. This allows the opening of branch cord bag **30** to be closed. Branch cord bag **30** is arranged to be wound around main cable **10**, making the coupling between branch cord bag **30** and main cable **10** stronger.

[0059] On the outer surface of main surface **31** where first hook or loop fastener portion **32***a* is provided in branch cord bag **30**, a fifth hook or loop fastener portion **32***e* is formed. In the present embodiment, fifth hook or loop fastener portion **32***e* forms a loop surface.

[0060] On the outer surface of main surface **31** where second hook or loop fastener portion **32***b* is provided in branch cord bag **30**, a sixth hook or loop fastener portion **32***f* is formed. In the present embodiment, sixth hook or loop fastener portion **32***f* forms a hook surface.

[0061] Fifth hook or loop fastener portion **32***e* is configured to be capable of coupling to sixth hook or loop fastener portion **32***f* of another branch cord bag **30** (refer to FIG. **13**). Fifth hook or loop fastener portion **32***e* and sixth hook or loop fastener portion **32***f* may be formed to extend in the longitudinal direction of main cable **10** (refer to FIG. **14**). Fifth hook or loop fastener portion **32***e* and sixth hook or loop fastener portion **32***f* are examples of the outer hook or loop fastener portion.

Bending Restriction Member

[0062] As illustrated in FIG. **2** and FIG. **5**, bending restriction member **40** is attached to a position corresponding to the branching portion of optical branch cable **1**. FIG. **7** is a perspective view of bending restriction member **40**. FIG. **8** illustrates bending restriction member **40** and branch cord portion **20**.

[0063] As illustrated in FIG. **7**, bending restriction member **40** includes a mounting portion **41**, a bending restriction portion **42**, and a main body portion **43**.

[0064] Mounting portion **41** is configured to be attachable to main cable **10**. In the present embodiment, bending restriction member **40** is provided with four mounting portions **41**. Bending restriction member **40** is attached to main cable **10** by sandwiching main cable **10** with four mounting portions **41**.

[0065] Main body portion **43** is formed in a curved shape so as to follow the outer shape of main cable **10**. Thus, main body portion **43** has a partially cylindrical shape.

[0066] Bending restriction portion **42** is provided to protrude from main body portion **43** in a direction intersecting the axial direction of main cable **10** when bending restriction member **40** is attached to main cable **10**. The boundary portion between bending restriction portion **42** and main body portion **43** is formed in a curved shape having a roundness so as to have a bending radius equal to or larger than a predetermined bending radius.

[0067] Bending restriction portion **42** restricts the position of branch cord portion **20** so that branch cord portion **20** is not bent with a radius less than the predetermined bending radius. As illustrated in FIG. **8**, branch cord portion **20** is led out from main cable **10** along the boundary portion between bending restriction portion **42** and main body portion **43**.

[0068] Main body portion **43** defines a connector pass-through portion **44** which is a hole through which optical connector **21** is able to pass, together with bending restriction portion **42**. When bending restriction member **40** is attached to main cable **10**, optical connector **21** and branch cord portion **20** are passed through connector pass-through portion **44** and accommodated in branch cord bag **30** together with bending restriction member **40**.

[0069] As shown in FIG. **2**, end portion bag **50** is provided to protect the end portion of optical cord

11 that exposed to outside at the end of main cable **10** and an end optical connector provided at the end portion of optical cord **11**.

[0070] Connecting loop **60** is connected to tension member **12** exposed at the end of main cable **10**. Connecting loop **60** is a portion connected to a pulling device such as a winch. Connecting loop **60** is formed in a ring shape, and when connecting loop **60** is pulled, tension is applied to tension member **12**, so that optical branch cable **1** can be displaced.

[0071] Next, a manufacturing process when branch cord portion **20** is led out from main cable **10** will be described.

[0072] In order to lead branch cord portion **20** out of main cable **10**, optical cord **11** needs to be exposed to the outside. In the present embodiment, tape **13** is partially peeled off from the overlapping portion of tape **13** which is longitudinally wrapped, so that optical cord **11** is easily exposed to the outside. In particular, when tape **13** has slit S formed therein, tape **13** can be easily peeled off by using slit S.

[0073] In particular, as illustrated in FIG. **4**, when slit S is formed from first end portion **13**A which is the end portion of the inner layer portion in the circumferential direction to second end portion **13**B which is the end portion of the outer layer portion in the circumferential direction as in the present embodiment, the work of exposing optical cord **11** to the outside is easy. Tape **13** is partially peeled off from second end portion **13**B so that first end portion **13**A is exposed to the outside, and when first end portion **13**A is exposed to the outside, tape **13** is partially peeled off from first end portion **13**A, thereby easily exposing optical cord **11** to the outside.

[0074] In the optical branch cable, if the outermost layer is formed by being covered with a resin by extrusion molding, it is necessary to remove the extrusion coating in order to lead out the branch cord portion from the main cable. At this time, it is necessary to remove the entire coating of the main cable in the circumferential direction in a portion of the cable longitudinal direction, and the time and labor required for removing the coating has been a factor of increasing the number of steps in manufacturing the optical branch cable.

[0075] In optical branch cable **1** according to the present embodiment, optical cord group **15** is bundled separately from tension member **12** by first bundle **14**, making it easy to work with and distinguish optical cord group **15** from tension member **12**. Thus, optical fiber core wire **11**A can be branched from optical cord group **15** to perform the work of manufacturing branch cord portion **20**. Tape **13** of the outermost layer covers core portion **17** by being wound by longitudinal wrapping. Thus, tape **13** can be partially peeled off from the overlapping portion of tape **13** which is longitudinally wrapped, and optical cord **11** can be exposed to the outside. Thus, optical fiber core wire **11**A can be branched from main cable **10** to perform the work of manufacturing branch cord portion **20**. Compared to a case where the outermost layer is formed by being coated with a resin by extrusion molding, it is not necessary to remove the extrusion coating when branch cord portion **20** is taken out, and thus it is possible to reduce the number of processes related to the manufacturing of optical branch cable **1** and to improve the ease of manufacturing optical branch cable **1**.

[0076] In optical branch cable **1** according to the present embodiment, tape **13** may be partially peeled off using slit S in order to expose optical cord **11** to the outside. Thus, it is easy to perform the manufacturing work of branch cord portion **20**. Further, since branch cord portion **20** branches from main cable **10** through slit S, the area where the inner portion of main cable **10** is exposed can be reduced after the manufacturing work of branch cord portion **20**.

[0077] In optical branch cable **1** according to the present embodiment, since first end portion **13**A which is the end portion of the inner layer portion in the circumferential direction and second end portion **13**B which is the end portion of the outer layer portion in the circumferential direction are partially removed by slit S, tape **13** can be easily partially peeled off from the end portion of tape **13** in the circumferential direction by using slit S.

[0078] In optical branch cable **1** according to the present embodiment, at least a portion of branch cord portion **20** is housed in branch cord bag **30** wound in a coil shape, making it possible to protect branch cord portion **20** in a state that is easy to take out after manufacturing.

[0079] In optical branch cable **1** according to the present embodiment, main surface **31** of branch cord bag **30** has a semicircular shape, making it easy to house in a state of being wound in a coil shape and fitted to branch cord portion **20**.

[0080] In optical branch cable **1** according to the present embodiment, branch cord bag **30** and main cable **10** are coupled by cable hook or loop fastener portion **18**, first hook or loop fastener portion **32***a*, and second hook or loop fastener portion **32***b*. Thus, branch cord bag **30** is less likely to be displaced with respect to main cable **10**, and thus branch cord portion **20** is less likely to lose its shape from the state of being wound in a coil shape.

[0081] In optical branch cable **1** according to the present embodiment, branch cord portion **20** is restricted by bending restriction portion **42** so as not to be bent with a radius less than the predetermined bending radius, and thus it is possible to suppress branch cord portion **20** from being damaged by unintended small-diameter bending.

[0082] It is noted that, optical cord group **15** is bundled separately from tension member **12** by first bundle **14**. This prevents tension member **12** from being entangled with optical cord **11** during the manufacturing process, the installation work, or the like. If tension member **12** becomes entangled with optical cord **11**, tension during installation work may be applied not only to tension member **12** but also to optical cord **11**. If the tension is applied to optical cord **11**, optical cord **11** may be damaged. In the present embodiment, first bundle **14** prevents tension member **12** from being entangled with optical cord **11**, and thus it is possible to prevent optical cord **11** from being damaged during the installation work.

[0083] Second bundle **16** bundles optical cord group **15** and tension member **12**. This can suppress differences in the relative positional relationship between optical cord group **15** and tension member **12** in the longitudinal direction of main cable **10**. In the longitudinal direction of main cable **10**, if the relative positional relationship between optical cord group **15** and tension member **12** is spiral, the tension applied to tension member **12** during installation may also be applied to optical cord group **15**. In the present embodiment, differences in the relative positional relationship between optical cord group **15** and tension member **12** in the longitudinal direction of main cable **10** can be suppressed, thereby reducing damage to optical cord **11** during installation work.

Pulling Jig

[0084] Next, a pulling jig **201** for pulling optical branch cable **1** will be described. FIG. **9** illustrates the installation of optical branch cable **1** using pulling jig **201**. As illustrated in FIG. **9**, a plurality of optical branch cables **1** are arranged in parallel in a state of being accommodated in a groove formed in cable laying tray T. Pulling jig **201** is used to pull optical branch cable **1** while maintaining a parallel state of the plurality of optical branch cables **1**. It is noted that the plurality of optical branch cables **1** are examples of the optical branch cable set. Further, pulling jig **201** and the optical branch cable set are examples of an optical cable pulling system.

[0085] FIG. **10** is a plan view of pulling jig **201**. As illustrated in FIG. **10**, pulling jig **201** includes a base portion **210**, a pulling portion **220**, and a plurality of connecting portions **230**. Base portion **210** has a width equal to or greater than a parallel state of the plurality of optical branch cables **1**. The “width equal to or greater than a parallel state of the plurality of optical branch cables **1**” is a width equal to or greater than the total sum of the diameters of the plurality of optical branch cables **1** in a parallel state. Base portion **210** may also have a width slightly less than the width of the groove formed in cable laying tray T, such that the base portion does not rotate in a top view relative to cable laying tray T illustrated in FIG. **9** during pulling.

[0086] As illustrated in FIG. **10**, guide portions each extending in a direction intersecting the width direction are formed in the width direction (left-right direction on the page) of base portion **210**. The number of the guide portions is the same as the number of connecting portions **230**. In the present embodiment, the guide portion is formed as a guide slit GS which is a groove formed in base portion **210**. Guide slit GS is formed to extend in a direction intersecting the width direction.

[0087] Base portion **210** includes a lower void space V below guide slit GS, and includes an engaged portion **211**, which defines an upper surface of void space V communicating with guide slit GS (refer to FIG. **11**). Void space V is formed along guide slit GS. Void space Vis formed to be wider than guide slit GS in the width direction.

[0088] Pulling portion **220** is provided to enabling a pulling device to be connected. In the present embodiment, pulling portion **220** is formed as a ring-shaped member to be hung on the pulling device.

[0089] The plurality of connecting portions **230** are provided at base portion **210** and arranged along the width direction (left-right direction on the page) of base portion **210**. Connecting portion **230**, by being hung with connecting loop **60**, can apply tension in the pulling direction to optical branch cable **1** when pulling jig **201** is pulled.

[0090] FIG. **11** is a cross-sectional view of the XI-XI plane in FIG. **10**. Connecting portion **230** includes a positioning portion **231** and an engaging tool **232**.

[0091] Positioning portion **231** includes a shaft portion **231***a* and an engaging portion **231***b*. Shaft portion **231***a* is able to be inserted through guide slit GS, and at least a portion of its outer surface is formed with external thread. Engaging portion **231***b* is a portion connected to shaft portion **231***a*. Engaging portion **231***b* is arranged below guide slit GS. Even when positioning portion **231** is displaced upward, engaging portion **231***b* abuts against engaged portion **211** and is configured not able to be inserted through guide slit GS.

[0092] Engaging tool **232** is provided with internal thread that is able to be fastened with shaft portion **231***a*. Engaging tool **232** is arranged on the outer surface of base portion **210** and is configured not able to be inserted through guide slit GS.

[0093] FIG. **12** illustrates the fixed positions of connecting portions **230**. Connecting portion **230** is displaceable along guide slit GS. Each of connecting portions **230** is fixable to any of at least two different positions in a direction intersecting the width direction. In the present embodiment, connecting portion **230** is fixable at a first position A on each guide slit GS or at a second position B closer to pulling portion **220** than first position A. It is note that, first position A and second position B are indicated by dashed line frames in FIG. **12**.

[0094] Referring back to FIG. **11**, connecting portion **230** is fixed by the fastening of engaging tool **232** and shaft portion **231***a*. By rotating engaging tool **232** in the direction where the fastening between engaging tool **232** and shaft portion **231***a* is tightened, positioning portion **231** displaces upward, causing engaging portion **231***b* to come into contact with engaged portion **211**. Since engaging tool **232** and shaft portion **231***a* sandwiches engaged portion **211**, connecting portion **230** is fixed to base portion **210** and cannot be displaced on guide slit GS.

[0095] The release of connecting portion **230** is performed by loosening the fastening between engaging tool **232** and shaft portion **231***a*. By rotating engaging tool **232** in the direction where the fastening between engaging tool **232** and shaft portion **231***a* is loosened, positioning portion **231** displaces below, causing engaging portion **231***b* to no longer come into contact with engaged portion **211**. When the sandwiching of engaged portion **211** by engaging tool **232** and shaft portion **231***a* is released, connecting portion **230** can be displaced from first position A to second position B or from second position B to first position A.

[0096] Accordingly, as illustrated in FIG. **12**, connecting portions **230** can be fixed at different positions along guide slit GS in a direction intersecting the width direction. Accordingly, the position at which connecting loop **60**, which is the end portion of optical branch cable **1**, is connected to pulling jig **201** can be adjusted for each optical branch cable **1**.

Optical Cable Pulling Method

[0097] Next, an optical cable pulling method will be described. The pulling of optical branch cable **1** is carried out while maintaining a parallel state in which the plurality of optical branch cables **1** are arranged in parallel in the width direction.

[0098] As illustrated in FIG. **12**, each of the plurality of connecting portions **230** is connected to one optical branch cable **1**. In this state, when pulling jig **201** is pulled, it is easier to apply the same direction and evenly distributed pulling force to all optical branch cables **1** connected to connecting portion **230**. Thus, all optical branch cables **1** are easily displaced in the same direction by the same amount, and the plurality of optical branch cables **1** can be pulled together while optical branch cables **1** are aligned.
[0099] FIG. **13** illustrates optical branch cable **1** connected to pulling jig **201**. FIG. **13** is a view of optical branch cable **1** as viewed of a horizontal direction intersecting the width direction. As illustrated in FIG. **13**, fifth hook or loop fastener portion **32***e* formed in branch cord bag **30** may be coupled to sixth hook or loop fastener portion **32***f* formed in branch cord bag **30** of adjacent optical branch cable **1**. Thus, branch cord bags **30** are coupled to each other, and optical branch cables **1** can be easily maintained in an aligned state during the pulling operation of optical branch cables **1**.
[0100] When all the optical branch cables are connected to one connecting portion, the optical branch cables are pulled toward one connecting portion. Thus, the direction and magnitude of the pulling force applied to the optical branch cable vary depending on the optical branch cable. In particular, since the optical branch cables arranged at the ends are pulled in the direction toward the center, the optical branch cables may run on or intersect the adjacent optical branch cables. As a result, the parallel state of the optical branch cables may not be maintained.
[0101] Pulling jig **201** and the optical cable pulling system according to the present embodiment include the plurality of connecting portions **230** arranged along the width direction. Each of connecting portions **230** is configured to enable one optical branch cable **1** to be connected. Thus, the plurality of optical branch cables **1** can be pulled in a state where each of optical branch cables **1** is connected to connecting portion **230**. This enables the plurality of optical branch cables **1** to be pulled together with as equal force as possible, thereby improving the installation workability of optical branch cables **1**.
[0102] Further, the optical cable pulling method according to the present embodiment includes pulling the plurality of optical branch cables **1** while maintaining a parallel state by using pulling jig **201**. By using pulling jig **201**, the plurality of optical branch cables **1** can be pulled simultaneously, and thus the installation workability of the optical cables are improved.
[0103] Meanwhile, a manufacturing error may occur in a position where a branching portion is formed, where branch cord portion **20** is led out from main cable **10**, for each optical branch cable **1**. Specifically, the distance from the end portion of optical branch cable **1** to one branching portion may be different between adjacent optical branch cables **1**. Further, the distance between the branching portions may be different between adjacent optical branch cables **1**. However, even when there is a manufacturing error, the positions of the branching portions need to be aligned to some extent in optical branch cables **1** arranged in parallel.
[0104] FIG. **14** illustrates optical branch cable **1** aligned by pulling jig **201**. It is noted that, in the illustration of FIG. **14**, the positions where branch cord bags **30** are provided indicates the positions of the branching portions in optical branch cable **1**. The branch cord bag closest to connecting loop **60** of the illustrated in FIG. **14** is referred to as a first branch cord bag **30**A, and the branch cord bag n-th closest to connecting loop **60** is referred to as an n-th branch cord bag.
[0105] As illustrated in FIG. **12**, connecting portion **230** is fixable at first position or second position different from each other along the cable longitudinal direction in each guide portion. At this time, when connecting portion **230** is connected to optical branch cable **1** having a relatively short distance from connecting loop **60** to first branch cord bag **30**A, connecting portion **230** is fixed to first position A. When connecting portion **230** is connected to optical branch cable **1** having a relatively long distance from connecting loop **60** to first branch cord bag **30**A, connecting portion **230** is fixed to second position B.
[0106] In this way, by ensuring that each connecting portion **230** is positioned at the appropriate first position A or second position B corresponding to each optical branch cable **1**, as illustrated in

FIG. **14**, it is possible to align the positions of first branch cord bags **30**A of optical branch cables **1** to some extent.
[0107] Further, the distance between branch cord bags **30** may be different depending on optical branch cable **1**. Even in such cases, by positioning each connecting portion **230** at first position A or second position B corresponding to optical branch cable **1**, as illustrated in FIG. **14**, it is possible to align the positions of n-th branch cord bags **30**N of optical branch cables **1** to some extent.
[0108] As described above, in pulling jig **201** according to the present embodiment, connecting portion **230** is fixable to any one of first position A and second position B which are two different positions in the direction intersecting the width direction. Thus, even when the position of the branching portion is displaced in the direction intersecting the width direction depending on optical branch cable **1**, connecting portion **230** can be shifted, and thus the position of the branching portion in the direction intersecting the width direction is easily adjusted to some extent. It is sufficient that each connecting portion **230** is fixable at two different positions in the direction intersecting the width direction, and the fixed positions in the direction intersecting the width direction in each connecting portion **230** may be the same as each other or may be different from each other. That is, a position of a first connecting portion **230** in the direction intersecting the width direction at first position A and a position of a second connecting portion **230** in the direction intersecting the width direction at first position A may be the same as each other or may be different from each other. Similarly, the position of the first connecting portion **230** in the direction intersecting the width direction at second position B and the position of the second connecting portion **230** in the direction intersecting the width direction at second position B may be the same as each other or may be different from each other.
[0109] In pulling jig **201** according to the present embodiment, connecting portion **230** is configured to be displaced along the guide portion, and thus a mechanism capable of changing the position of connecting portion **230** in the direction intersecting the width direction can be realized by a simple configuration.
[0110] In pulling jig **201** according to the present embodiment, engaging tool **232** is fastened or loosened with respect to shaft portion **231***a*, and thus it is possible to realize a mechanism for displacing connecting portion **230** with respect to base portion **210** in the direction intersecting the width direction and fixing the position with a simple configuration.
[0111] In optical branch cable **1** according to the present embodiment, branch cord bag **30** is provided with fifth hook or loop fastener portion **32***e* and sixth hook or loop fastener portion **32***f*, which are outer hook or loop fastener portions. In the optical branch cable set according to the present embodiment, in a state where the optical branch cables are arranged in parallel, branch cord bags **30** of adjacent optical branch cables **1** are coupled to each other by fifth hook or loop fastener portion **32***e* and sixth hook or loop fastener portion **32***f*. Thus, when the optical branch cable set is pulled, the plurality of optical branch cables **1** can move integrally. As described above, according to the optical branch cable set of the present embodiment, it is easy to manufacture the optical branch cable set for installing the plurality of optical branch cables **1** in a collective and parallel state.
[0112] Even when the positions of first branch cord bags **30**A are aligned, the positions of n-th branch cord bags **30**N are not necessarily aligned in the width direction (that is, completely adjacent). As illustrated in FIG. **14**, the positions of n-th branch cord bags **30**N may be shifted in the longitudinal direction of optical branch cables **1**. However, in the optical branch cable set according to the present embodiment, fifth hook or loop fastener portion **32***e* and sixth hook or loop fastener portion **32***f* extend along the longitudinal direction of main cable **10**. Thus, even when the branching portions are shifted in the longitudinal direction of optical branch cable **1**, branch cord bags **30** are easily coupled to each other.
Modification of Optical Branch Cable Set
[0113] Next, modification of the optical branch cable set will be described. FIG. **15** is a perspective

view of an optical branch cable set **300** according to the modification. FIG. **16** illustrates optical branch cable set **300** and a branch cord bag **330** according to the modification.

[0114] Optical branch cable set **300** according to the modification includes 22 optical branch cables **301**. Optical branch cable **301** according to the modification is different from optical branch cable **1** according to the embodiment shown in FIG. **9** in that the plurality of optical branch cables **301** share one branch cord bag **330**. In other words, 22 branch cord portions **320** are accommodated in one branch cord bag **330**.

[0115] FIG. **17** illustrates an arrangement example of branch cord portion **320** of each optical branch cable **301**. For reference, optical branch cables **301** are referred to as a first optical branch cable, a second optical branch cable, . . . , and a twenty-second optical branch cable in the order of arrangement from left to right on the page of FIG. **17**. Similarly, branch cord portions **320** of the first optical branch cable, the second optical branch cable, . . . , and the twenty-second optical branch cable are referred to as a first branch cord portion **320**A, a second branch cord portion **320**B . . . , and a twenty-second branch cord portion **320**V, respectively.

[0116] As illustrated in FIG. **17**, branch cord portions **320** are arranged in three rows in the longitudinal direction of optical branch cable **301**. In the upper row, branch cord portions **320** from first branch cord portion **320**A to a seventh branch cord portion **320**G are arranged. In the center row, branch cord portions **320** from an eighth branch cord portion **320**H to a fourteenth branch cord portion **320**N are arranged. In the lower row, branch cord portions **320** from a fifteenth branch cord portion **320**O to twenty-second branch cord portion **320**V are arranged. In plan view, each branch cord portion **320** is wound clockwise in a coil shape.

[0117] Branch cord portion **320** may be arranged so that the bending radius is as large as possible. For example, in the upper row, first branch cord portion **320**A is arranged at a position farthest from a branching portion BR. The branch cord portions are arranged in order from first branch cord portion **320**A to seventh branch cord portion **320**G, at positions away from branching portion BR. In other words, seventh branch cord portion **320**G is arranged at a position closest to branching portion BR.

[0118] Similarly, in the center row, eighth branch cord portion **320**H is arranged at a position farthest from branching portion BR. Eighth branch cord portion **320**H to fourteenth branch cord portion **320**N are arranged in this order at positions away from branching portion BR. In other words, fourteenth branch cord portion **320**N is arranged at a position closest to branching portion BR. Further, in the lower row, fifteenth branch cord portion **320**O is arranged at a position farthest from branching portion BR. Fifteenth branch cord portion **320**O to twenty-second branch cord portion **320**V are arranged in this order at positions away from branching portion BR. In other words, twenty-second branch cord portion **320**V is arranged at a position closest to branching portion BR.

[0119] As branch cord portions **320** are arranged in this way, branch cord portion **320** arranged at the position closest to branching portion BR is branch cord portion **320** of rightmost optical branch cable **301** in each row. At this time, when branch cord portion **320** gets wound clockwise in a coil shape, branch cord portion **320** arranged at the position closest to branching portion BR is less likely to be applied to an excessive small-diameter bending load.

[0120] Further, when branch cord portion **320** gets taken out, branch cord portions **320** are arranged in order, and thus, when branch cord portion **320** is connected to server rack **120** or the like, branch cord portions **320** are hardly mistaken, and the connection work of branch cord portion **320** is easy.

[0121] In optical branch cable **301** according to the modification, since one branch cord bag **330** accommodates all branch cord portions **320**, when optical branch cable set **300** is pulled, the plurality of optical branch cables **301** are easily moved integrally.

[0122] Although the present disclosure has been described in detail with reference to the specific embodiments, it will be apparent to those skilled in the art that various changes and modifications can be made without departing from the spirit and scope of the present disclosure. The number, positions, shapes, and the like of the constituent members described above are not limited to those in the above embodiments, and can be changed to the number, positions, shapes, and the like suitable for carrying out the present disclosure.

[0123] In the present embodiment, the case of the installation of optical branch cable **1** using pulling jig **201** has been illustrated and explained, but the cable installed using pulling jig **201** is not limited to optical branch cable **1**. For example, an optical cable without a branch cord portion may be pulled by using a pulling jig while maintaining a parallel state.

[0124] In the present embodiment, connecting portion **230** is fixable at first position A and second position B on guide slit GS, but connecting portion **230** is fixable at three or more points on guide slit GS, and each connecting portion **230** is fixable at, at least two or more different positions on guide slit GS.

Claims

1. A pulling jig for pulling a plurality of optical cables while maintaining a state in which the optical cables are arranged in parallel, the pulling jig comprising: a base portion having a width equal to or more than a width in the state in which the plurality of optical cables are arranged in parallel; a pulling portion for enabling the base portion to be connected to a pulling device; and a plurality of connecting portions provided at the base portion and arranged in a width direction along the width, wherein the connecting portions are connectable to the optical cables in at least one-to-one correspondence.

2. The pulling jig according to claim 1, wherein each of the optical cables includes a main cable including a plurality of optical fiber core wires, and a branch cable portion in which at least one of the optical fiber core wires is led out from the main cable, and wherein the connecting portions are fixable at any one of at least two different positions in a direction intersecting the width direction.

3. The pulling jig according to claim 2, wherein a guide portion extending in the direction intersecting the width direction is formed at the base portion, and wherein each of the connecting portions is displaceable along the guide portion.

4. The pulling jig according to claim 3, wherein the guide portion is formed as a guide slit, wherein each of the connecting portions includes a positioning portion and an engaging tool, wherein the positioning portion includes a shaft portion being able to be inserted through the guide slit, an external thread being formed on at least a portion of an outer surface, and an engaging portion disposed below the guide slit, the engaging portion being not able to be inserted through the guide slit, wherein an internal thread being able to be fastened to the shaft portion is formed in the engaging tool, and wherein a position of the connecting portion is fixed with respect to the base portion in the direction intersecting the width direction by the engaging tool being fastened to the shaft portion.

5. An optical cable pulling method comprising pulling a plurality of optical cables while maintaining a state in which the optical cables are arranged in parallel using the pulling jig according to claim 1.

6. An optical cable pulling system comprising: the pulling jig according to claim 1; and an optical cable set including a plurality of optical cables, wherein each of the optical cables includes a main cable including a plurality of optical fiber core wires, and a branch cable portion in which at least one of the optical fiber core wires is branched from the main cable, wherein the plurality of optical cables are arranged in parallel in the width direction, and wherein the connecting portions are connected to the optical cables in one-to-one correspondence.